

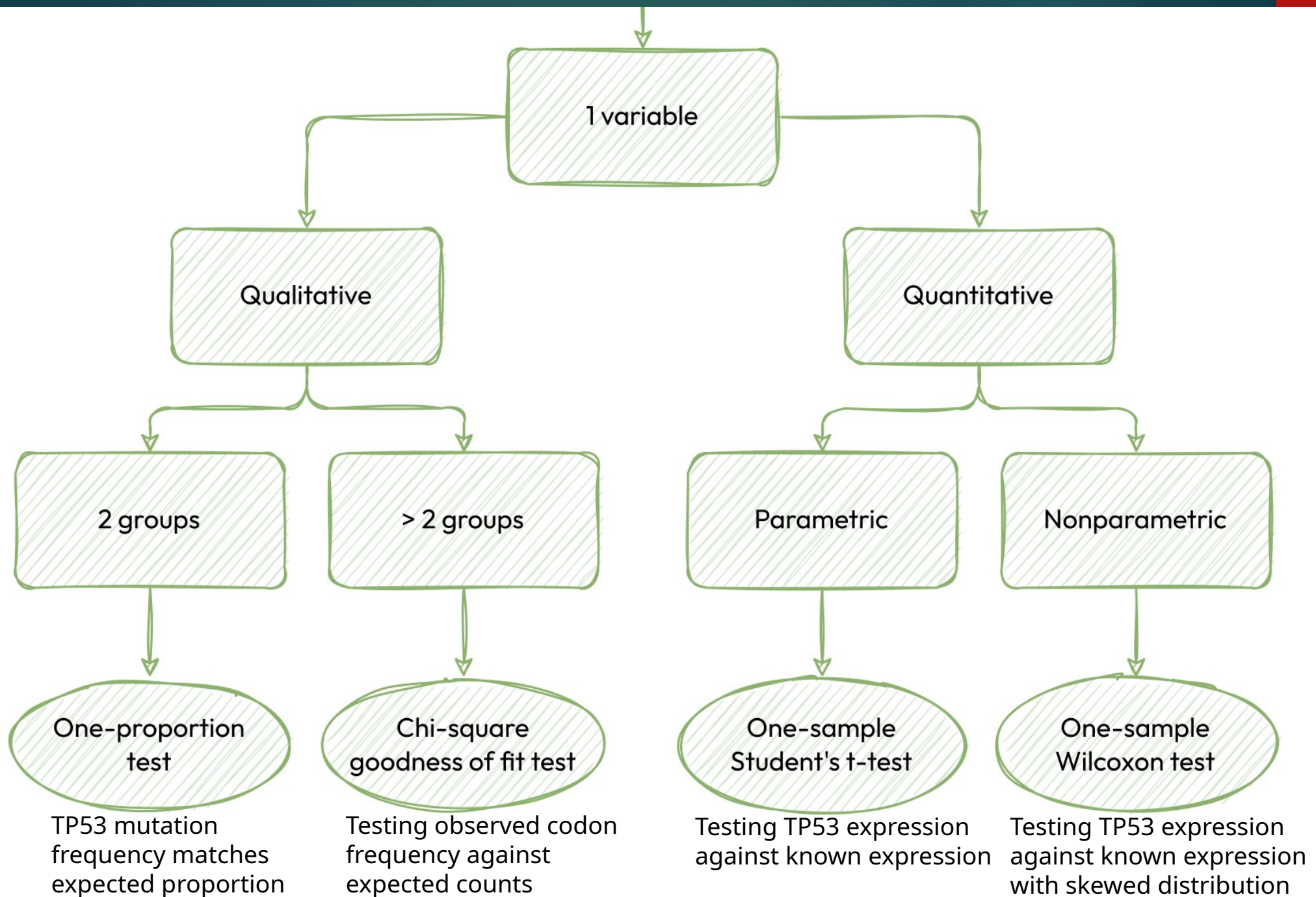


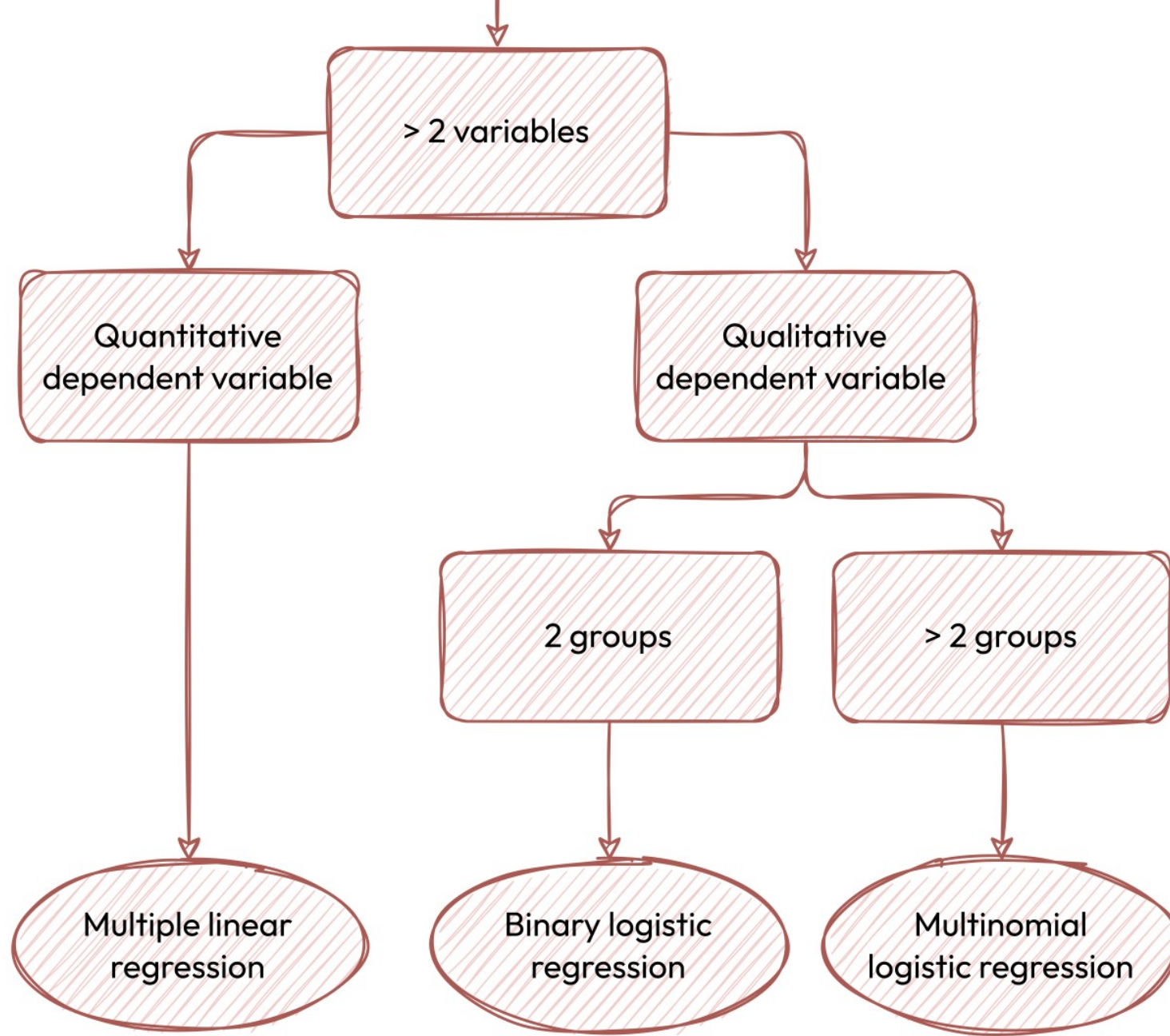
Biostats

JAMES ROUSE

Power calculations

- ▶ Power is the probability of correctly rejecting the null hypothesis when the alternative hypothesis is true.
- ▶ Power calculations can tell you the number of samples that you need to achieve a certain level of statistical power.
- ▶ Mean difference, significance level, and power can be set to calculate the minimum number of samples required.
- ▶ Example: How many samples needed to achieve 1.5 log fold change difference between tumor and normal gene expression at 80% power and 0.05 significance level





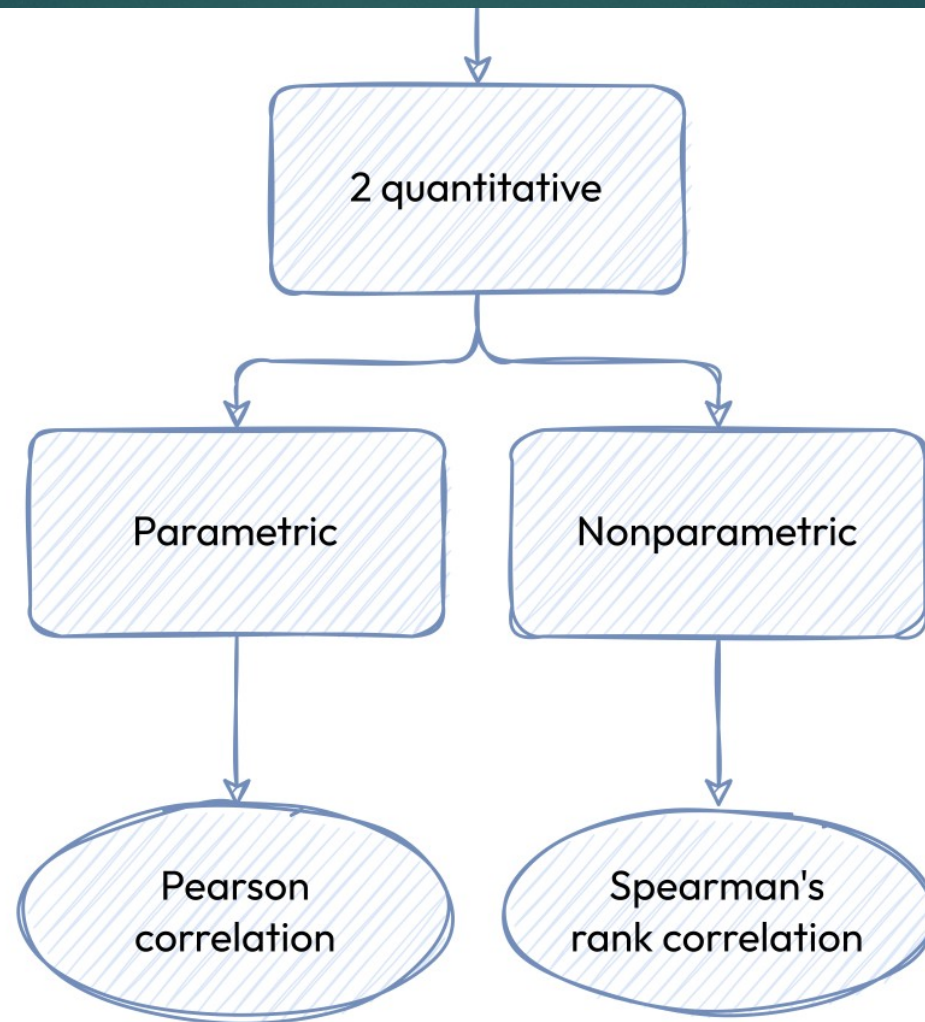
Predicting tumor size based on expression of 4 genes

Predicting tumor presence based on gene expression

Predicting tumor stage based on gene expression

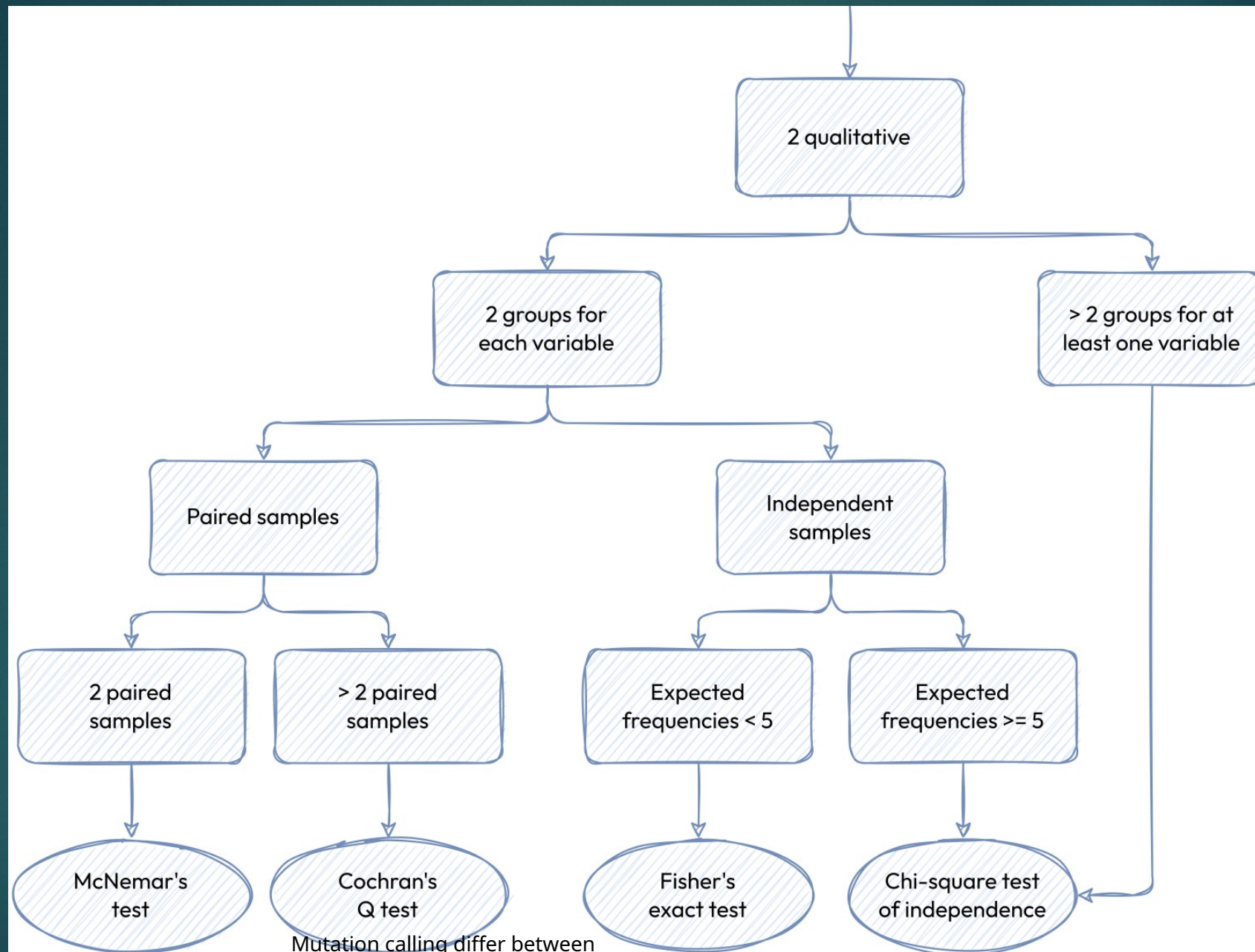
Regression

- ▶ Linear Regression models the relationship between a continuous outcome variable and one or more continuous or categorical predictor variables.
- ▶ Logistic Regression predicts the probability of a binary outcome (e.g., disease vs. no disease) based on predictor variables.



Co-expression with linear relationship

Co-expression with non-linear relationship



Co-expression with
linear relationship

Mutation calling differ between
different variant callers?

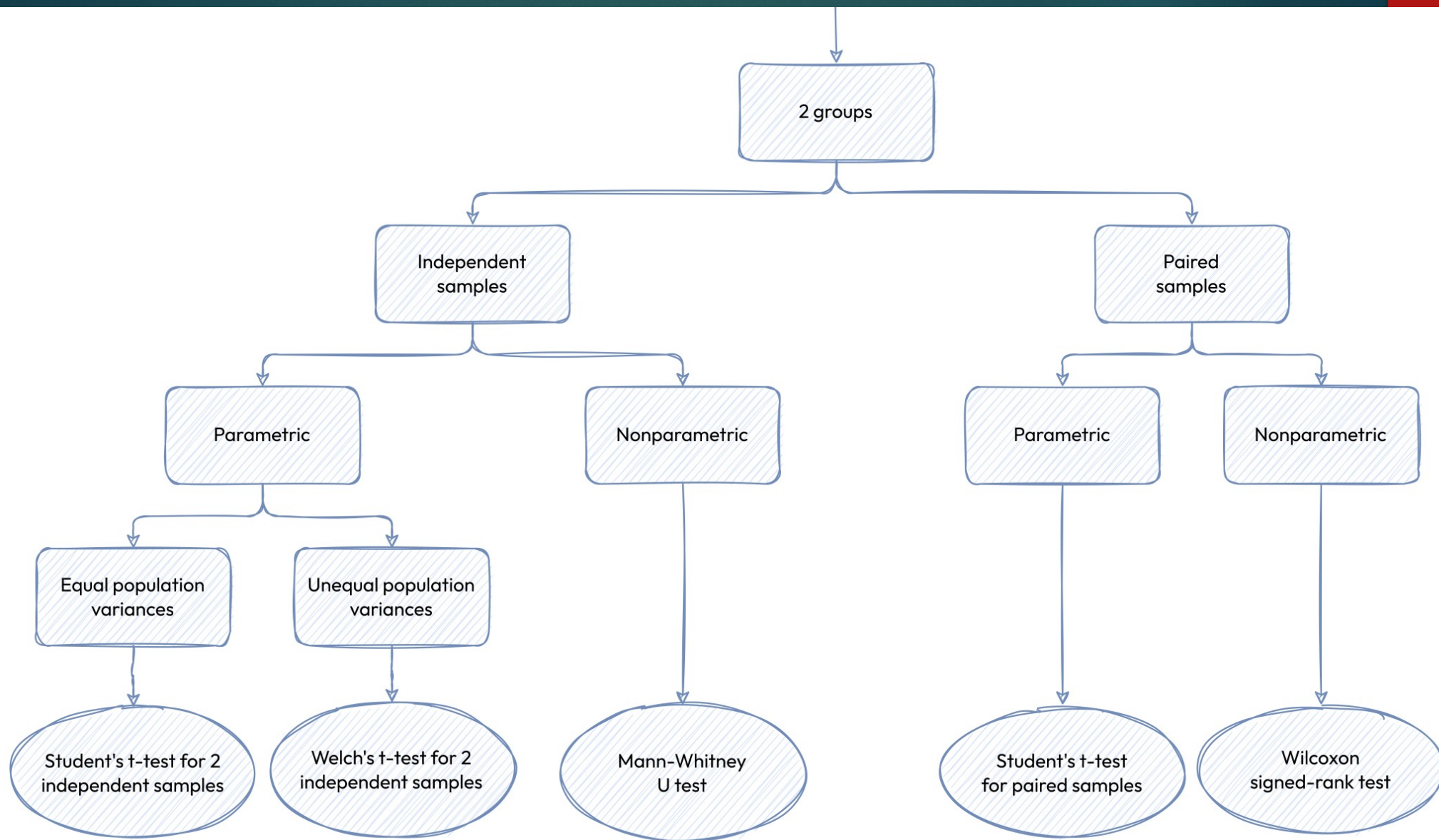
Test whether presence of a
mutation is associated with drug
response

Chi-Square test

- ▶ Chi-Squared is used to test whether there is an association between two categorical variables
- ▶ For example, the presence of a gene mutation (yes or no) and disease status.
- ▶ Assumptions: Categorical data, independent observations, at least 5 samples in each category
- ▶ Calculation involves looking at observed and expected in each category
- ▶ Goodness of fit test used when you have expected values, independence test used when you want to see if two categorical variables are associated with each other

Fisher's Exact Test

- ▶ Fisher's Exact Test is preferred when expected counts in any contingency table cell are < 5 , where the Chi-Square test may be inaccurate.
- ▶ Exact Probability Calculation: Unlike Chi-Square, which uses an approximation, Fisher's test calculates the exact p-value based on all possible contingency tables.
- ▶ Works for Small 2x2 Tables: Ideal for small datasets (e.g., mutation presence vs. disease status in rare conditions).
- ▶ Chi-Square is Better for Large Datasets: If sample sizes are large, Chi-Square is computationally faster and provides a reliable approximation.



TP53 expression in tumor vs normal samples

TP53 expression in tumor vs normal samples (skewed distribution)

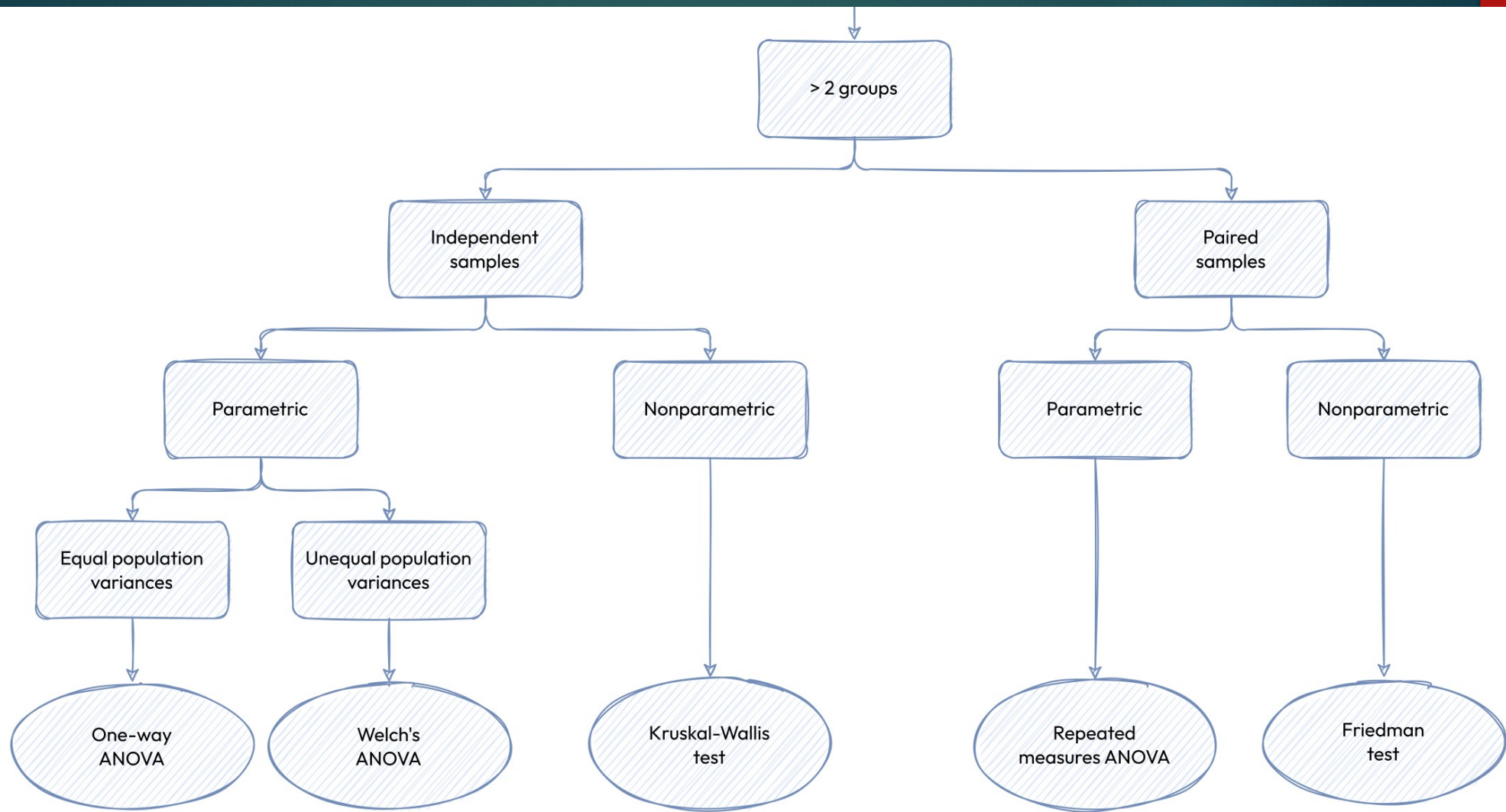
TP53 expression in tumor vs adjacent normal samples

T-test

- ▶ Compares the means of two different groups.
- ▶ Paired and unpaired
 - ▶ Paired example – comparing same sample before and after treatment
 - ▶ Unpaired – Tumor vs normal
- ▶ T-test assumes data follows a normal distribution and that the samples have equal variance
- ▶ Bioinformatics example – Whether VIM expression is different cell types

Mann-Whitney U Test/Wilcoxon

- ▶ Non-parametric test comparing two independent groups when data is skewed or non-normal.
- ▶ **Assumptions:** Groups are independent, the response variable is continuous, and distributions should have similar shapes.
- ▶ **Method:** Ranks all values (or differences in values if paired), sums ranks for each group, computes a test statistic, and converts it to a p-value.
- ▶ **Application:** Used in bioinformatics to compare gene expression or protein levels between two conditions, often times when the distribution is skewed.
- ▶ Kruskal-Wallis test is a non-parametric ranked test used for three or more groups



Gene expression across 3 conditions

TP53 expression in metastatic vs tumor vs adjacent normal samples

ANOVA

- ▶ ANOVA is used to compare the means of three or more groups rather than just two groups
- ▶ One-way ANOVA – used to compare means across one variable (gene expression across 3 conditions)
- ▶ Two-way ANOVA – used to compare means across two variables (gene expression across 3 conditions and time points)
- ▶ Measures the between group and within group variance to see if the variance between groups is significantly higher (different).
- ▶ If there is significance, a post-hoc test can tell you which groups differ

MANOVA

- ▶ MANOVA tests whether the combination of dependent variables differs across groups
- ▶ Example: 3 treatment groups and 15 genes with expression values. MANOVA will test whether the expression patterns are different across groups
- ▶ If MANOVA is significant, at least one of the groups differs and post-hoc tests like an individual ANOVA is needed

Correlation of continuous variables

- ▶ Pearson correlation – Used for linear relationship between two variables.
 - ▶ Assumes that the relationship between the two variables is linear and that the data is normal.
- ▶ Spearman correlation – Used for monotonic relationships (doesn't need linear assumption). Non-parametric correlation of continuous variables